

# 1.2: Complex Numbers

## 1. The Imaginary Unit and Complex Numbers

**Imaginary unit:**  $i$  is defined so that  $i^2 = -1$ , giving  $i = \sqrt{-1}$ .

**Complex number:** a number of the form  $z = a + bi$  where  $a, b \in \mathbb{R}$ . The set of all complex numbers is denoted  $\mathbb{C}$ .

**Real part:**  $\text{Re}(z) = a$ . **Imaginary part:**  $\text{Im}(z) = b$  (the *real* coefficient of  $i$ , not  $bi$ ).

Every real number is a complex number with  $\text{Im}(z) = 0$ ; if  $a = 0$  and  $b \neq 0$ , the number  $bi$  is called *purely imaginary*.

### Powers of $i$

The powers of  $i$  cycle with period 4:

| $n$   | $i^1$ | $i^2$ | $i^3$ | $i^4$ | $i^5$ | $i^6$ |
|-------|-------|-------|-------|-------|-------|-------|
| $i^n$ | $i$   | $-1$  | $-i$  | $1$   | $i$   | $-1$  |

For  $i^n$ : divide  $n$  by 4 and use the remainder —  $0 \rightarrow 1, 1 \rightarrow i, 2 \rightarrow -1, 3 \rightarrow -i$ .

## 2. Arithmetic of Complex Numbers

### Addition and Subtraction

$$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i$$

## Multiplication

Expand using FOIL then substitute  $i^2 = -1$ :

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

## Complex Conjugate

The **complex conjugate** of  $z = a + bi$  is  $z^* = \bar{z} = a - bi$ .

Key properties:

- $z + \bar{z} = 2a = 2 \operatorname{Re}(z)$  (real)
- $z - \bar{z} = 2bi = 2i \operatorname{Im}(z)$  (purely imaginary)
- $z\bar{z} = a^2 + b^2 = |z|^2$  (real and non-negative)
- $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$  and  $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$

## Division

Rationalise the denominator by multiplying numerator and denominator by the conjugate of the denominator:

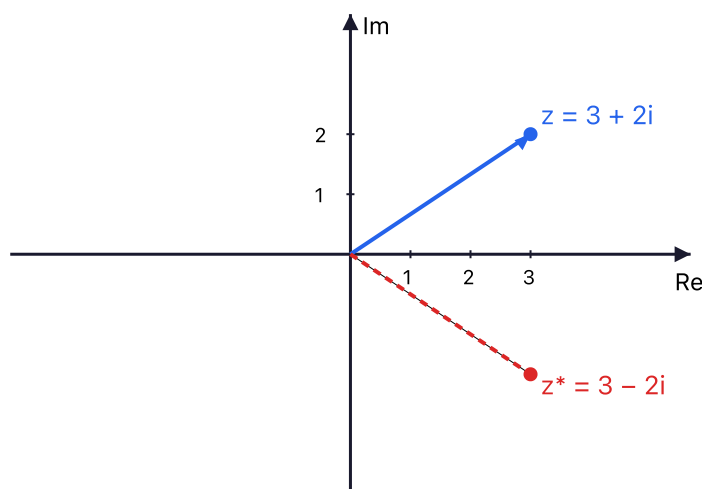
$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}$$

**Worked example:** Simplify  $\frac{3 + 2i}{1 - i}$ .

$$\text{Multiply by } \frac{1 + i}{1 + i}: \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)} = \frac{3 + 3i + 2i + 2i^2}{1 - i^2} = \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i.$$

## 3. The Argand Diagram

Complex numbers are represented as points (or vectors from the origin) in the plane, plotting  $\text{Re}(z)$  horizontally and  $\text{Im}(z)$  vertically.



## Modulus and Argument

**Modulus:**  $|z| = \sqrt{a^2 + b^2}$  — the distance from the origin to the point  $(a, b)$ .

**Argument:**  $\arg z = \theta$  — the angle from the positive real axis, measured anticlockwise. The *principal argument* is the value with  $-\pi < \theta \leq \pi$ .

**OCR convention:** OCR uses the principal argument in the range  $-\pi < \arg z \leq \pi$ . Always state your range and check the quadrant — using  $\arctan(b/a)$  without thinking gives the wrong angle in quadrants 2 and 3.

| Quadrant         | Signs          | Argument formula          |
|------------------|----------------|---------------------------|
| 1 (top-right)    | $a > 0, b > 0$ | $\arctan(b/a)$            |
| 2 (top-left)     | $a < 0, b > 0$ | $\pi + \arctan(b/a)$      |
| 3 (bottom-left)  | $a < 0, b < 0$ | $-\pi + \arctan(b/a)$     |
| 4 (bottom-right) | $a > 0, b < 0$ | $\arctan(b/a)$ (negative) |

## 4. Modulus-Argument and Exponential Form

$$z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$$

where  $r = |z|$  and  $\theta = \arg z$ . The exponential form follows from **Euler's formula**:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

**Convert  $z = -1 + i\sqrt{3}$  to mod-arg form.**

$|z| = \sqrt{1+3} = 2$ . Second quadrant, so  $\arg z = \pi - \arctan(\sqrt{3}/1) = \pi - \pi/3 = 2\pi/3$ .

So  $z = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) = 2e^{i2\pi/3}$ .

## Multiplication and Division in Polar/Exponential Form

Multiplication adds arguments, division subtracts them:

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

- $|z_1 z_2| = |z_1| |z_2|$ ,  $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
- $|z_1 / z_2| = |z_1| / |z_2|$ ,  $\arg(z_1 / z_2) = \arg z_1 - \arg z_2$

**OCR Exam Tip:** Exponential form is usually the cleanest for multiplication, division and powers. Convert to mod-arg form first, then apply the rule, then convert back if needed.

## 5. De Moivre's Theorem

**De Moivre's theorem:** For any  $n \in \mathbb{Z}$ ,

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

Equivalently,  $(e^{i\theta})^n = e^{in\theta}$ .

This is proved by induction for  $n \geq 0$ , and the result for negative  $n$  follows using  $(e^{i\theta})^{-1} = e^{-i\theta}$ .

## Uses of de Moivre

1. **Powers of complex numbers:**  $z^n = r^n e^{in\theta}$ .
2. **Multiple-angle identities:** expand  $(\cos \theta + i \sin \theta)^n$  by the binomial theorem, then equate real and imaginary parts.
3.  **$n$ th roots:** see §7.

**Show that**  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ .

By de Moivre,  $\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3$ .

Expand:  $(c + is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = c^3 - 3cs^2 + i(3c^2s - s^3)$ .

Real part:  $\cos 3\theta = c^3 - 3cs^2 = c^3 - 3c(1 - c^2) = 4c^3 - 3c$ . ■

## Using $z + 1/z$ and $z - 1/z$

If  $z = e^{i\theta}$  then:

$$z + z^{-1} = 2 \cos \theta, \quad z - z^{-1} = 2i \sin \theta$$

$$z^n + z^{-n} = 2 \cos(n\theta), \quad z^n - z^{-n} = 2i \sin(n\theta)$$

These are powerful for expressing  $\cos^n \theta$  or  $\sin^n \theta$  as a sum of cosines/sines of multiple angles (needed for integration).

**Express  $\cos^4 \theta$  in terms of  $\cos 4\theta$ ,  $\cos 2\theta$ , 1.**

$$\begin{aligned} (2 \cos \theta)^4 &= (z + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4} \\ &= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6 = 2 \cos 4\theta + 8 \cos 2\theta + 6. \end{aligned}$$

$$\text{So } 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6 \implies \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}.$$

## 6. Roots of Polynomial Equations

**Fundamental Theorem of Algebra:** every polynomial of degree  $n \geq 1$  with complex coefficients has exactly  $n$  roots in  $\mathbb{C}$  (counting multiplicity).

**Conjugate root theorem:** if a polynomial has *real* coefficients and  $z = p + qi$  is a root, then  $\bar{z} = p - qi$  is also a root. Complex roots always occur in conjugate pairs.

**Given that  $2 + i$  is a root of  $z^3 - 7z^2 + 17z - 15 = 0$ , find the other roots.**

By the conjugate root theorem,  $2 - i$  is also a root. The quadratic factor is  $(z - (2 + i))(z - (2 - i)) = z^2 - 4z + 5$ .

Divide:  $z^3 - 7z^2 + 17z - 15 = (z^2 - 4z + 5)(z - 3)$ . So the third root is  $z = 3$ .

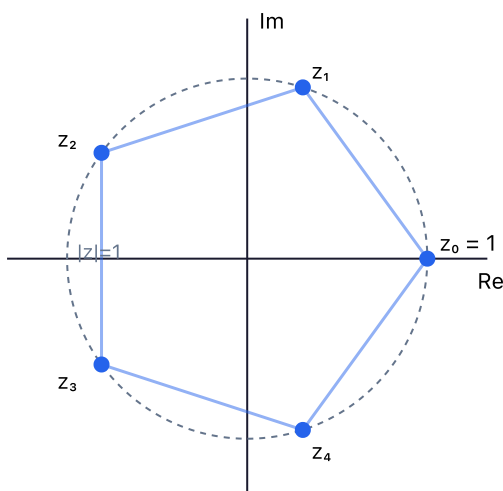
## 7. Roots of Unity and $n$ th Roots

The  $n$ th **roots of unity** are the solutions of  $z^n = 1$ . There are exactly  $n$  of them, given by

$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, 2, \dots, n-1.$$

On an Argand diagram, the  $n$ th roots of unity lie on the unit circle at the vertices of a regular  $n$ -gon, one vertex at  $z = 1$ .

The fifth roots of unity — vertices of a regular pentagon on the unit circle.



## $n$ th Roots of a General Complex Number

To solve  $z^n = w$  where  $w = R e^{i\phi}$ :

$$z_k = R^{1/n} e^{i(\phi+2\pi k)/n}, \quad k = 0, 1, \dots, n-1$$

All  $n$  roots have the same modulus  $R^{1/n}$  and arguments differing by  $2\pi/n$  — they lie equally spaced on a circle of radius  $R^{1/n}$ .

**Find all solutions of  $z^4 = -16$ .**

Write  $-16 = 16 e^{i\pi}$ . Then  $z_k = 2 e^{i(\pi+2\pi k)/4} = 2 e^{i\pi(1+2k)/4}$  for  $k = 0, 1, 2, 3$ .

Arguments:  $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$  (or equivalently  $-3\pi/4, -\pi/4$ ).

So  $z = \sqrt{2}(1 \pm i), \sqrt{2}(-1 \pm i)$ .

**OCR Exam Tip:** Always sketch the roots on an Argand diagram — it prevents sign errors and makes the symmetry obvious. Quote arguments in the principal range  $-\pi < \theta \leq \pi$  unless the question asks otherwise.

## Sum of Roots of Unity

For  $n \geq 2$ , the sum of all  $n$ th roots of unity is zero:

$$\sum_{k=0}^{n-1} e^{2\pi i k/n} = 0$$

Proof: this is a geometric series with ratio  $e^{2\pi i/n} \neq 1$ , sum  $= \frac{1 - e^{2\pi i}}{1 - e^{2\pi i/n}} = 0$ .

## 8. Loci in the Argand Diagram

Let  $z = x + iy$  and interpret  $|z - w|$  as the distance between  $z$  and  $w$  in the plane.

| Locus         | Shape                           | Cartesian form                                                    |
|---------------|---------------------------------|-------------------------------------------------------------------|
| $ z - a  = r$ | Circle, centre $a$ , radius $r$ | $(x - \operatorname{Re} a)^2 + (y - \operatorname{Im} a)^2 = r^2$ |

|                                |                                                       |                          |
|--------------------------------|-------------------------------------------------------|--------------------------|
| $ z - a  =  z - b $            | Perpendicular bisector of segment $ab$                | Linear equation          |
| $\arg(z - a) = \alpha$         | Half-line from $a$ (excluding $a$ ) at angle $\alpha$ | Linear ray               |
| $ z - a  = k z - b , k \neq 1$ | Circle of Apollonius                                  | Circle (after expanding) |

**Sketch the locus  $|z - 2 + i| = 3$  and find its Cartesian equation.**

Rewrite:  $|z - (2 - i)| = 3$ . Circle, centre  $(2, -1)$ , radius 3.

Cartesian:  $(x - 2)^2 + (y + 1)^2 = 9$ .

**Find the Cartesian equation of  $|z - 3| = |z + i|$ .**

Distance from  $(3, 0)$  equals distance from  $(0, -1)$  — the perpendicular bisector of the segment joining these two points.

Let  $z = x + iy$ :  $(x - 3)^2 + y^2 = x^2 + (y + 1)^2$ .

Expand:  $-6x + 9 = 2y + 1 \implies y = 4 - 3x$ .

**OCR Exam Tip:** For loci questions always (a) sketch first to visualise, (b) then do the algebra. Note whether the locus is a line, circle, ray, or region — and whether endpoints are included (strict vs. non-strict inequality).

## 9. Summary of Formulae

| Quantity         | Formula                              |
|------------------|--------------------------------------|
| Modulus          | $ z  = \sqrt{a^2 + b^2}$             |
| Conjugate        | $z^* = a - bi$                       |
| Mod-arg form     | $z = r(\cos \theta + i \sin \theta)$ |
| Exponential form | $z = r e^{i\theta}$                  |

|                              |                                                                   |
|------------------------------|-------------------------------------------------------------------|
| Euler's formula              | $e^{i\theta} = \cos \theta + i \sin \theta$                       |
| De Moivre                    | $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ |
| $n$ th roots of unity        | $e^{2\pi i k/n}, k = 0, \dots, n-1$                               |
| $n$ th roots of $Re^{i\phi}$ | $R^{1/n} e^{i(\phi+2\pi k)/n}$                                    |

## 10. Exam Style Strategy

- **Algebra in Cartesian form** — best for addition, subtraction, basic multiplication/division.
- **Mod-arg / exponential form** — best for powers, roots, multiple-angle identities.
- **Argand diagrams** — always sketch when loci, roots of unity or multiple roots are involved.
- **Check your principal argument range** and remember the quadrant rules.
- **Conjugate pairs** — if a real-coefficient polynomial has one complex root, you get a second for free.